

Edge-Deployed Maritime Perception for Unmanned Surface Vehicles

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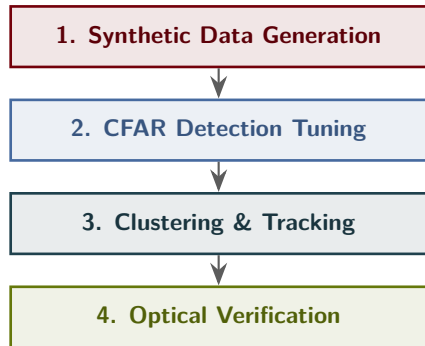
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Overview

Four papers addressing the full **perception pipeline** for autonomous surface vessels using COTS sensors and edge hardware:

- 1 Synthetic Radar Data** Vaught
Multi-mode simulation for training data generation
- 2 CFAR Autotuning** Kirk
Human-supervised detection parameter optimization
- 3 Adaptive ST-DBSCAN** Cancilla
Real-time clustering and multi-sweep tracking
- 4 Dual-Camera Acquisition** Wang
Active optical verification and auto-labeling



Shared Sensor Platform & Field Sites

Common Hardware

- **Radar:** Furuno DRS4D NXT X-band
 - 9.38–9.44 GHz, solid-state, 100 W
 - LFM chirp, ~ 20 m range resolution
 - 3.9° beamwidth, 24/36/48 RPM
- **Cameras:** 2 $\times$ FLIR M300C (wide + PTZ)
- **Edge compute:**
 - NVIDIA Jetson AGX Orin (32 GB)
 - Raspberry Pi 5 (browser-based tools)
- **Battery:** 12.8 V / 50 Ah LiFePO₄

Field Collection Sites (South Carolina)

- Lake Murray – primary test site
- Lake Greenwood – inland variety
- Charleston Harbor – coastal/maritime

~ 30 sessions over 11 months

Targets: recreational boats, pontoons, buoys

Target Characteristics

- Fiberglass boats: $\sigma = 0.5\text{--}20\text{ m}^2$
- Aluminum / buoys: $\sigma = 1\text{--}50\text{ m}^2$
- Antenna height: 2–4 m above water
- Grazing angle: $\approx 0.19^\circ$ at 500 m

Paper 1

A Multi-Mode Simulation Framework
for X-Band Pulse Compression Radar

J.C. Vaught, D. Cahl, Y. Wang

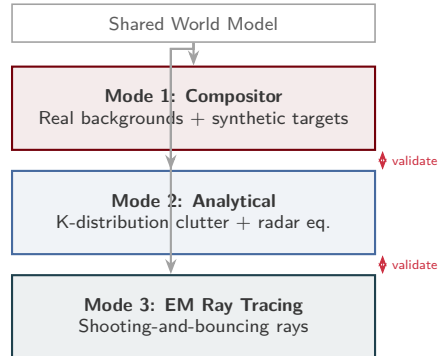
Synthetic Radar – Motivation

The Problem

- Maritime radar data is **scarce** and expensive
- Manual labeling of radar returns is ambiguous
- Coverage of rare events is limited
- Domain shift between environments

Three-Mode Synthetic Framework

- **Mode 1:** Data-driven compositing (fast)
- **Mode 2:** Analytical clutter generation
- **Mode 3:** EM ray tracing (physics valid.)



Synthetic Radar – Framework Details

1. Background selection
2. Target signature generation
3. Amplitude matching
4. Insertion & blending
5. Automatic label generation

Augmentation: Real radar interference
and rain clutter overlays

Mode 2: Analytical

K-distribution clutter PDF:

$$f(x) = \frac{2}{\Gamma(\nu)} \left(\frac{\nu}{P_c} \right)^{\frac{\nu+1}{2}} x^{\nu} K_{\nu-1} \left(2\sqrt{\frac{\nu x}{P_c}} \right)$$

- Shape ν controls tail weight
- Swerling-I target model
- Full B-scan via radar equation

Mode 3: Ray Tracing

- Shooting-and-bouncing rays
- Time-varying water mesh from wave spectrum
- Physics validation of Modes 1–2
- Shoreline geometry (ongoing)

Radar Eq:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L}$$

Domain Transfer: YOLO detector trained on Mode 1 composites, evaluated on real radar frames

Synthetic Radar – Sensor Characterization

Furuno DRS4D NXT Specifications

Parameter	Value
Transmitter	Solid-state
Frequency	9.38–9.44 GHz
Output Power	100 W
Modulation	LFM chirp
Pulse Width	5–18 μ s
Range Res.	~20 m
Beamwidth	3.9°
Rotation	24/36/48 RPM
PRF	700–1100 Hz
Azimuth Quant.	8192 spokes/rev
Radome Dia.	610 mm

Data Acquisition

- UDP multicast capture at spoke rate
- B-scan format: range $\times$ azimuth
- 30+ collection sessions, 3 SC sites
- 11-month campaign

Key Contributions

1. Multi-mode framework with cross-validation
2. Sim-to-real domain transfer demonstration
3. Environmental augmentation strategy
4. First DRS4D NXT characterization for DL

Paper 2

Human-Supervised CFAR Autotuning
for Maritime Sensing

N. Kirk, J.C. Vaught, D. Cahl, Y. Wang

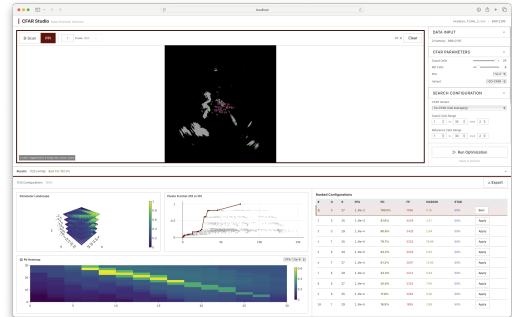
CFAR Autotuning – CFAR Studio

The Problem

- CFAR detectors need tuning: guard cells (N_g), reference cells (N_r), P_{FA}
- Heavy-tailed clutter invalidates Gaussian assumptions
- Manual tuning is slow and not reproducible

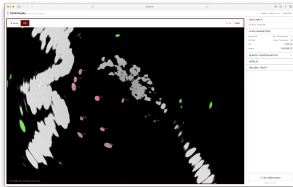
CFAR Studio – Open-Source Tool

- React 19 + TypeScript, fully client-side
- Grid search via Web Worker
- Integral images: $O(1)$ noise estimation
- **~500 configs in <10 seconds**



CFAR Studio: PPI with optimization results dashboard

CFAR Autotuning – Detection & Scoring



PPI: CFAR

detections (green) vs. ground truth (pink)



B-scan with CFAR

detection overlay

Scoring Function

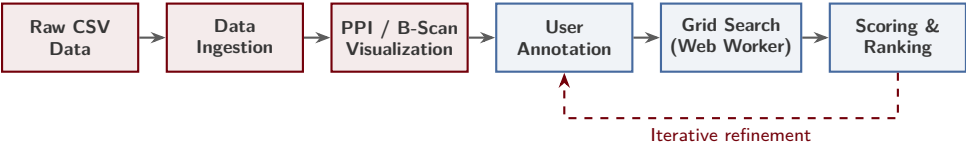
Coverage ratio and frame score:

$$\rho_k = \frac{\sum_{p \in B_k} D_t(p)}{|B_k|} \quad S = 0.7 \min_k s_k + 0.3 \bar{s}$$

Key Findings

- Guard cells $N_g \geq 3$: better extended targets
- Optimal reference: $N_r \in [10, 20]$
- Top configs: stability > 0.9
- **GO-CFAR**: fewest FA near clutter
- **SO-CFAR**: highest detection, more FA
- **CA-CFAR**: best in homogeneous scenes

CFAR Autotuning – Processing Pipeline



CA-CFAR

$$Z^{CA} = \frac{1}{N} (S_{out} - S_{in})$$

Average of all ref. cells

GO-CFAR

$$Z^{GO} = \max(\bar{Z}_1, \bar{Z}_2)$$

Greatest-of two halves

SO-CFAR

$$Z^{SO} = \min(\bar{Z}_1, \bar{Z}_2)$$

Smallest-of two halves

Detection: $D_t(i, j) = 1[l_t(i, j) > \alpha \cdot Z_{i,j}] \quad \alpha = N(P_{FA}^{-1/N} - 1)$

Paper 3

Surface-Object Detection and Tracking
Using Adaptive ST-DBSCAN

S. Cancilla, J.C. Vaught, D. Cahl, Y. Wang

Adaptive ST-DBSCAN – Approach

Pipeline



Key Innovations

- **Adaptive ε :** k-NN distance statistics

$$\varepsilon_s = \max \left(P_{90} \{ d_k \}, \varepsilon_{\min} \right)$$

- **Gap recovery:** Union-Find merging

$$d_{\max} = v_{\max} \cdot \Delta t \cdot T_s$$

- **Hungarian assignment:** optimal track association

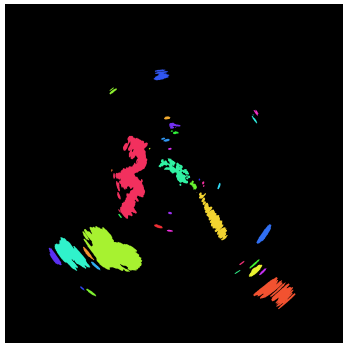
Implementation

Rust + Rayon (data-parallel k-NN)

Spatial hashing: $h = 2\varepsilon_{\max}$

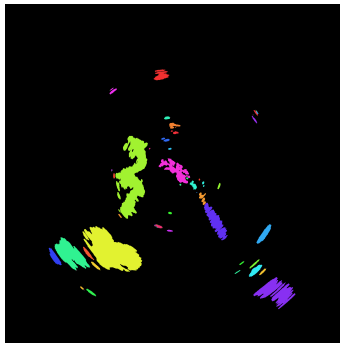
Target: NVIDIA Jetson AGX Orin

Adaptive ST-DBSCAN – Clustering Comparison



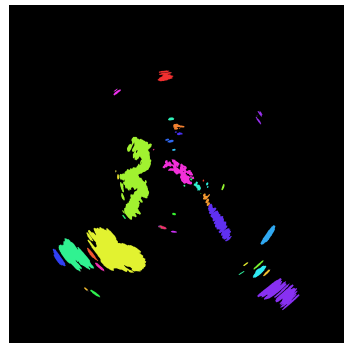
Vanilla DBSCAN

1693 clusters (over-segmented)



Adaptive ST-DBSCAN

1177 clusters (reduced noise)

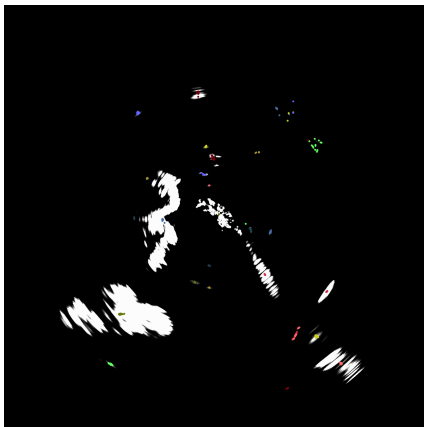


+ Gap Recovery

Consolidated track fragments

Synthetic X-band radar data – compound K-distribution clutter with Bernoulli flicker model (15–40% dropout)

Adaptive ST-DBSCAN – Tracking Results



Final tracking: 3 objects with stable IDs

MOT Metrics Comparison

Method	MOTA	MOTP	Recall	IDSW
Vanilla DBSCAN	-6.97	1.80	0.987	18
Std. ST-DBSCAN	-0.10	—	0.000	0
Adaptive ST-DBSCAN	-4.44	1.19	0.974	18
+ Gap Recovery	-3.82	1.18	0.812	0

Edge Performance (Jetson AGX Orin)

Stage	Time (ms)
Adaptive ε (k-NN)	196
ST-DBSCAN clustering	169
Gap recovery + extraction	53
Hungarian + I/O	50
Total	511 $\pm$ 93

~ 2 sweeps/s > real-time for 48 RPM radar

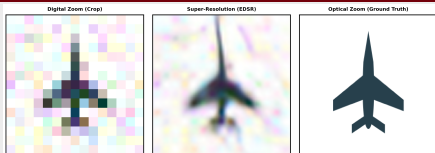
Paper 4

Dual-Camera Active Acquisition
for Small-Object Dataset Construction

J. Wang, J.C. Vaught, D. Cahl, Y. Wang

Dual-Camera Acquisition – Concept

Digital Zoom vs. Super-Resolution vs. Optical Zoom



Left: pixelated crop |

Center: SR hallucination | Right: optical zoom

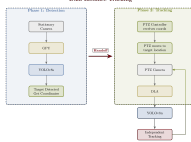
Key Idea

Replace human labelers with a **dual-camera system**:

- **Wide-angle** (fixed): persistent detection coverage
- **PTZ** (steerable): optical zoom verification
- Hard small-object detection → trivial classification

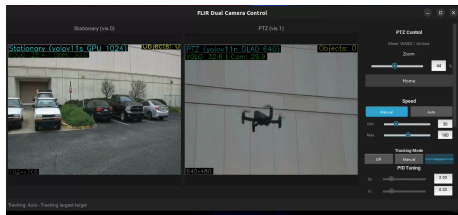


2× FLIR M300C on 0.9 m stand



Dual-instance tracking pipeline

Dual-Camera Acquisition – Drone Tracking Demo



Wide-angle: $\sim 20 \times 15$ px target (GPU, YOLOv11s)



PTZ

zoom: $\sim 120 \times 90$ px (DLA, YOLOv11n) – $6\times$ resolution

Detector Throughput

	Wide (GPU)	PTZ (DLA)
Model	YOLOv11s	YOLOv11n
Inference	27.9 ms	46.5 ms
Throughput	35.9 FPS	21.5 FPS
Detection	100%	99%

Aggregate: **57.4 FPS** (true parallel)

Tracking Controller

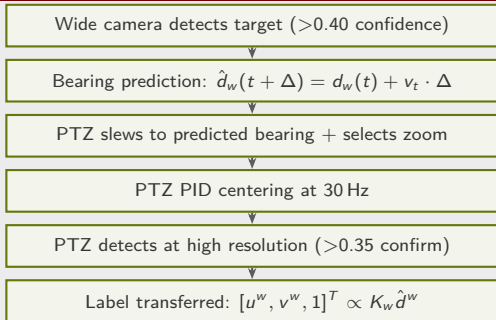
- PID: $K_p = 1.5$, $K_i = 0.05$, $K_d = 1.0$ at 30 Hz
- 2% centering deadzone
- MJPEG saves 60–130 ms vs. H.264

42-Second Drone Flight

Continuous track, no reacquisition needed

Dual-Camera Acquisition – Automated Labeling

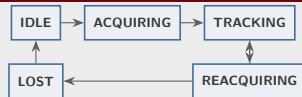
Slew-to-Classification Pipeline



Quality Gates

- PTZ confirm threshold: 0.35
- Class agreement required
- Motion blur rejection
- Exposure validation
- Boundary rejection

Five-State Machine



Summary

Unified Perception Pipeline

Vaught: Synthetic training data (3 modes)



Kirk: CFAR parameter optimization



Cancilla: Real-time tracking (511 ms/sweep)



Wang: Optical verification (57.4 FPS)

Key Takeaways

- All COTS sensors, no custom hardware
- Edge-deployable on Jetson / RPi5
- Real-time performance demonstrated
- Open-source tools (CFAR Studio)
- Field-validated at 3 SC sites

Future Work

- Complete YOLO domain-transfer evaluation
- Kalman filtering + MHT for tracking
- AIS fusion for ground truth
- Multi-PTZ scheduling
- Full system integration test

Questions?

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